

COMP0078: Supervised Learning

Collected Problem Sheet

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Contents

1	Introduction to Supervised Learning	2
2	Features, Kernels and Regularisation	2
3	Support Vector Machines	4
4	Tree-based Learning and Ensemble Methods	4
5	Online Learning	5
6	Graphs in Machine Learning	6
7	Learning Theory	7

1 Introduction to Supervised Learning

Problem 1.1. “For the K -nearest neighbour algorithm, larger K values will tend to lead to overfitting.” True or False? Explain your answer.

Problem 1.2. First, give Cover’s bound on the Bayes error of the 1-nearest neighbour algorithm as the number of examples goes to infinity. Second, give an example where the 1-nearest neighbour algorithm achieves the Bayes error as the number of examples goes to infinity, and explain your reasoning.

Problem 1.3. How can we choose k to ensure that the k -NN makes an unambiguous decision in the two-class case, provided the test point is not equidistant to any pair of points in the training set? What values of k give an unambiguous decision under similar assumptions for the 3-class case?

Problem 1.4. One of the drawbacks of the nearest-neighbour algorithm is that we must retain all of the training data. Describe a situation where a training point can be removed without affecting the resulting 1-NN classification for any test point in the input space.

Problem 1.5. In linear regression it is common to transform the training data

$$(x_1, y_1), \dots, (x_m, y_m)$$

to

$$((x_1, 1), y_1), \dots, ((x_m, 1), y_m),$$

i.e., we add an additional component to each input vector and set it to 1. What is the motivation for this procedure?

Problem 1.6. Given a dataset $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$ where $x \in \mathbb{R}^n$ and $y \in \{1, 2, \dots, K\}$. How can linear regression (least squares) be used or adapted to tackle such a multi-class classification problem?

Problem 1.7. Given $Y \subset \mathbb{R}$ and $|Y| < \infty$ and $|X| < \infty$ and a probability mass function $p : X \times Y \rightarrow \mathbb{R}$, find

$$f^* := \arg \min_f \mathbb{E}[|x - f(y)|]$$

(Bayes estimator with L_1 loss).

2 Features, Kernels and Regularisation

Problem 2.1. If f and g are convex then $f + g$ is convex.

Problem 2.2. If f is convex and g is affine (linear + a constant) then $f(g(\cdot))$ is convex.

Problem 2.3. Suppose M is a symmetric matrix. Then M is a PSD matrix if and only if $f(x) = x^\top M x$ is convex.

Problem 2.4. A level set of a convex function is convex.

Problem 2.5. For $f : X \rightarrow \mathbb{R}$, if $X \subseteq \mathbb{R}^n$ is a convex set and if the Hessian of f evaluated at x , denoted $H(x)$, is positive semidefinite for all $x \in X$, then f is convex.

Problem 2.6. Consider the solution to the linear regression optimisation problem. When is it advantageous to compute it via the primal solution? When is it advantageous to compute it via the dual solution? Explain why.

Problem 2.7. Given a kernel $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $K(x, t) := (1 + \langle x, t \rangle)^2$. Find a feature map $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ which corresponds to the kernel.

Problem 2.8. For each of the following functions $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$, argue whether they are a valid kernel and when the answer is positive derive an associated feature map representation.

(a) $K(x, t) = x^\top D t$ where $D = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$

(b) $K(x, t) = x^\top D t$ where $D = \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$

(c) $K(x, t) = \exp(x_1 t_1)$

(d) $K(x, t) = x^\top t - (x^\top t)^2$

(e) Prove that if $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a valid kernel then the following are also valid:

(i) $K(Ax, At)$ where A is a given 2×2 matrix.

(ii) $f(x)K(x, t)f(t)$ where f is a given real-valued function.

Problem 2.9. Consider a Gaussian kernel function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $K(x, z) := e^{-\|x-z\|^2}$. Does there exist a finite-dimensional feature map representation? I.e., does there exist a $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^d$ such that $K(x, z) = \langle \varphi(x), \varphi(z) \rangle$? Indicate “yes” or “no” and provide an argument. [Hard]

Problem 2.10. Argue that the vector space definition implies that every element $x \in X$ has an additive inverse $-x \in X$ such that $x + (-x) = 0$.

Problem 2.11. Kernels between sets. Let X be a finite set. Define $K : 2^X \times 2^X \rightarrow \mathbb{R}$ as

$$K(A, B) := 2^{|A \cap B|}$$

where $A, B \subseteq X$. Prove that K is a kernel.

Problem 2.12. Min kernel.

(a) Argue that $\min(x, t)$ (where $x, t \in [0, \infty)$) is a kernel. [Technical]

(b) Let $M > 0$. Find a set of basis functions $\varphi_\ell : [0, M] \rightarrow \mathbb{R}$ for $\ell \in \mathbb{N}$ such that

$$\min(x, t) = \sum_{\ell=0}^{\infty} \varphi_\ell(x) \varphi_\ell(t).$$

[Technical, very difficult]

3 Support Vector Machines

Problem 3.1. Describe the criterion used by hard margin Support Vector Machines to choose a separating hyperplane for a linearly separable dataset, illustrating with a diagram indicating the margin and the support vectors.

Problem 3.2. For a hard-margin SVM, if we remove one of:

- (a) the examples which is a support vector from the training set,
- (b) the examples which is not a support vector from the training set,

and retrain without that example, how does the maximum margin change?

Problem 3.3. Consider the optimisation given by

$$\min_{w, \xi} \|w\| + C \sum_{i=1}^m \xi_i$$

subject to $y_i \langle w, x_i \rangle \geq 1 - \xi_i$, $\xi_i \geq 0$, $i = 1, \dots, m$.

Which part of the optimisation corresponds to the regulariser? What is the loss function incorporated into the optimisation?

Problem 3.4. Assume that the set $S = \{(x_i, y_i)\}_{i=1}^m \subset \mathbb{R}^2 \times \{-1, 1\}$ is strictly linearly separable by a line going through the origin. Consider the optimisation problem:

$$P_1 : \min_w \frac{1}{2} w^\top w \quad \text{subject to } y_i w^\top x_i \geq 1, \quad i = 1, \dots, m.$$

Argue that the above problem has a unique solution. Describe the geometric meaning of this solution.

Problem 3.5. Show that the vector w solving problem P_1 has the form $w = \sum_{i=1}^m c_i y_i x_i$ where $c_1, \dots, c_m$ are some nonnegative coefficients. [Hint: use the method of Lagrange multipliers]

Problem 3.6. Show that the coefficients $c_1, \dots, c_m$ solve the optimisation problem

$$P_2 : \max_{c \geq 0} -\frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m c_i c_j y_i y_j x_i^\top x_j + \sum_{i=1}^m c_i.$$

Finally, if $(c_1, \dots, c_m)$ is the solution to this problem and w is the solution to P_1 , argue that $w^\top w = \sum_{i=1}^m c_i$.

4 Tree-based Learning and Ensemble Methods

Problem 4.1. Given the dataset

$$\{((1, 1), 9), ((1, 2), -4), ((1, 3), 2), ((2, 2), 4), ((2, 3), 2)\},$$

find the regression tree corresponding to the greedy recursive partitioning procedure with respect to square error.

Problem 4.2. Explain how a random forest is constructed. What is the motivation for the tree construction method?

Problem 4.3. Both bagging and boosting produce predictions by creating an ensemble of functions. Explain how these ensembles differ (you do not need to describe the methods for arriving at the ensembles, just how structurally the ensembles will differ).

Problem 4.4. The Adaboost initialisation and update.

- (a) Explain how Adaboost chooses its initial weighting $D_0(i)$ of the examples $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$.
- (b) Explain how it updates the distribution D_t after choosing a weak learner h_t with coefficient α_t at stage t , including the normalising constant Z_t .
- (c) Give an intuitive justification for this update.

Problem 4.5. Explain how Adaboost uses the distribution D_t that weights the examples to choose a weak learner from a set H at stage t .

Problem 4.6. Using the results of Problem 4.4(b), obtain a bound for

$$\frac{1}{m} \sum_{i=1}^m \mathbb{I}[\text{sign}(f_T(x_i)) \neq y_i] \leq \frac{1}{m} \sum_{i=1}^m \exp(-y_i f_T(x_i)),$$

in terms of the normalisation constants $Z_1, \dots, Z_T$, where $f_T(x) = \sum_{t=1}^T \alpha_t h_t(x)$.

Problem 4.7. A dataset $(x_1, y_1), \dots, (x_m, y_m)$ is generic iff $x_i = x_j \implies y_i = y_j$. A set of functions H is called weakly-universal if for every generic dataset and weighting $\{D_i\}_{i=1}^m$ with $\sum_{i=1}^m D_i = 1$, there exists $h \in H$ with weighted error $\sum_{i=1}^m D_i [h(x_i) \neq y_i] < 0.5$.

- (a) A positive decision stump $h_{i,z} : \mathbb{R}^n \rightarrow \{-1, 1\}$ is defined as

$$h_{i,z}(x) := \begin{cases} 1 & \text{if } x_i \leq z, \\ -1 & \text{if } x_i > z. \end{cases}$$

The set of all decision stumps is $H_{ds} := \{s h_{i,z} : s \in \{-1, 1\}, i \in \{1, \dots, n\}, z \in \mathbb{R}\}$. Is H_{ds} weakly-universal? Provide an argument.

- (b) Let S denote the set of all strings of non-zero finite length over $\{a, b\}$. A positive substring-match function $g_z : S \rightarrow \{-1, 1\}$ is defined as

$$g_z(x) := \begin{cases} 1 & \text{if } x \sqsubseteq z, \\ -1 & \text{if } x \not\sqsubseteq z. \end{cases}$$

The set of all substring-match functions is $H_{ss} := \{s g_z : s \in \{-1, 1\}, z \in S\}$. Is H_{ss} weakly-universal? Provide an argument.

5 Online Learning

Problem 5.1. Suppose $X = \{\text{True}, \text{False}\}^n$. Give a polynomial time algorithm A with a mistake bound $M(A) \leq O(n^2)$ for any example sequence which is consistent with a k -literal conjunction. Your answer should contain an argument that $M(A) \leq O(n^2)$.

Problem 5.2. State the perceptron convergence theorem (Novikoff) explaining the relation with the hard margin support vector machine solution.

Problem 5.3. Define the c -regret of a learning algorithm as

$$c\text{-regret}(m) = L_A(S) - c \min_{i \in [n]} L_i(S).$$

- (a) For the weighted majority set-up, argue that without randomised prediction it is impossible for all training sequences to obtain sublinear c -regret for $c < 2$.
- (b) Show how to select β to obtain sublinear 2-regret.

Problem 5.4. Consider binary prediction with expert advice, with a perfect expert. Prove that any algorithm makes at least $\Omega(\min(T, \log_2 n))$ mistakes in the worst case.

Problem 5.5. Recall that by tuning the weighted majority we achieved a bound

$$M \leq 2.63 \min_i M_i + 2.63 \ln n.$$

Now, by using randomisation in the prediction, design an algorithm with a bound that has the property

$$\frac{M}{T} \leq \frac{\min_{i \in [n]} M_i}{T}$$

as $T \rightarrow \infty$, for the weighted majority setting.

6 Graphs in Machine Learning

Problem 6.1. Describe, in brief, the differences between semi-supervised learning, supervised learning and unsupervised learning.

Problem 6.2. Given the unlabeled data points $(-2, 0), (-1, -1), (0, 0), (1, 1), (2, 0)$, draw the 2-nearest neighbour graph with respect to the Euclidean distance.

Problem 6.3. Give the graph Laplacian for the dataset above.

Problem 6.4. In the labelled graph (a path $v_1 - v_2 - v_3 - v_4 - v_5$ with v_1 labelled $+1$ and v_5 labelled -1):

- (a) What is the value of the minimum cut solution at vertex v_3 ?
- (b) What is the value of the harmonic energy solution at vertex v_3 ?

Problem 6.5. Consider the parameterised graph D_m consisting of two m -cliques connected by a single bridge edge. The first clique contains a node labelled -1 and the second contains a node labelled $+1$. Consider the labelling of D_m as resulting from harmonic energy minimisation. What is the labelling in the limit as $m \rightarrow \infty$? Justify your answer.

Problem 6.6. [Hard] Prove that

$$\min_{u \in \{-1, 1\}^n} \left\{ \sum_{ij} w_{ij} |u_i - u_j| \mid u_{i_1} = y_1, \dots, u_{i_\ell} = y_\ell \right\} = \min_{u \in \mathbb{R}^n} \left\{ \sum_{ij} w_{ij} |u_i - u_j| \mid u_{i_1} = y_1, \dots, u_{i_\ell} = y_\ell \right\}.$$

Problem 6.7. [Hard] Prove the following inequality for the resistance diameter:

$$\max_{k \in [n]} L_{kk}^+ \leq \max_{i, j \in [n]} r_G(i, j).$$

Problem 6.8. [Hard] Prove the equality

$$L_{vv}^+ = \frac{R(v)}{n} - \frac{R_{\text{tot}}}{n^2}.$$

Problem 6.9. [Hard] For an unweighted connected graph, prove

$$\sum_{(i,j) \in G} r_G(i,j) = n - 1.$$

Problem 6.10. [Hard] Let L be a graph Laplacian for an n -vertex graph. Consider the kernel $K := (L + \varepsilon I)^{-1}$ where $\varepsilon > 0$. Prove that for any $x \in [n]^\ell$ and any $y \in \{-1, 1\}^\ell$,

$$\arg \min_{\substack{u \in \mathbb{R}^n \\ u_{x_1}=y_1, \dots, u_{x_\ell}=y_\ell}} u^\top K^{-1} u = \arg \min_{\substack{u \in \mathbb{R}^n \\ u_{x_1}y_1 \geq 1, \dots, u_{x_\ell}y_\ell \geq 1}} u^\top K^{-1} u.$$

What does this imply for “support vectors” in Laplacian-based kernels? Argue that there exists a positive definite $n \times n$ matrix K for which this property does not hold.

7 Learning Theory

Problem 7.1. Let $X = \mathbb{R}^2$, $Y = \{0, 1\}$, and let H be the class of concentric circles in the plane, i.e., $H = \{h_r : r > 0\}$ where $h_r(x) = \mathbf{1}(\|x\|_2 \leq r)$. Prove that H is PAC learnable (assume realizability), and its sample complexity is bounded by

$$m_H(\varepsilon, \delta) \leq \left\lceil \frac{\ln(1/\delta)}{\varepsilon} \right\rceil.$$

Problem 7.2. Prove $1 - x \leq e^{-x}$ for $x \in \mathbb{R}$.

Problem 7.3. Prove that if $|X| \geq km$ for a positive integer $k \geq 2$, then we can replace the lower bound of $1/4$ in the No-Free-Lunch theorem with $\frac{k-1}{2k}$. Namely, let A be a learning algorithm for binary classification. Let m be any number smaller than $|X|/k$. Then, there exists a distribution D over $X \times \{0, 1\}$ such that:

- There exists $f : X \rightarrow \{0, 1\}$ with $L_D(f) = 0$.
- $\mathbb{E}_{S \sim D^m} [L_D(A(S))] \geq \frac{1}{2} - \frac{1}{2k}$.

Problem 7.4. Given a finite domain set X and a number $k \leq |X|$, determine the VC-dimension of each of the following classes:

- $H_{=k}^X = \{h \in \{0, 1\}^X : |\{x : h(x) = 1\}| = k\}$.
- $H_{\text{at-most-}k}^X = \{h \in \{0, 1\}^X : |\{x : h(x) = 1\}| \leq k \text{ or } |\{x : h(x) = 0\}| \leq k\}$.

Problem 7.5. We have shown that for a finite hypothesis class H , $\text{VCdim}(H) \leq \lfloor \ln(|H|) \rfloor$. However, the VC-dimension can be much lower:

- (a) Find an example of a class H of functions over $X = [0, 1]$ such that H is infinite while $\text{VCdim}(H) = 1$.
- (b) Give an example of a finite hypothesis class H over $X = [0, 1]$ where $\text{VCdim}(H) = \lfloor \log_2(|H|) \rfloor$.

Problem 7.6. On page 49 (Learning Theory slides), why was the VC-dimension of large margin classifiers formulated with respect to a fixed input space X rather than for example $\{x \in \ell_2 : \|x\|_2 \leq R\}$?

Problem 7.7. Given $X \subset \ell_2$ and $\Lambda \in (0, \infty)$, define

$$H_{X,\Lambda} = \{x \mapsto \text{sign}(\langle w, x \rangle) : w \in \ell_2, \|w\|_2 \leq \Lambda, |\langle w, x \rangle| \geq 1 \forall x \in X\}.$$

- (a) Prove that $\text{VCdim}(H_{X,\Lambda}) \leq \Lambda^2 \max_{x \in X} \|x\|_2^2$.
- (b) Show that for every $\Lambda \in (0, \infty)$ there exists an X such that $\text{VCdim}(H_{X,\Lambda}) = \lfloor \Lambda^2 \max_{x \in X} \|x\|_2^2 \rfloor$.